

The future

Design and validation engineers need to power their DUTs safely and easily during manual tests or automated sequences. They are frequently under pressure to perform tests quickly, and their test benches are often crowded.

"Our customers told us they want a Keysight-quality bench power supply with modern I/O, a compact footprint and excellent bench usability," says Kari Fauber, general manager, Keysight's Power and Energy Division. "We built the E36100 series to meet the needs of R&D and design validation engineers who face increasing pressures to produce high-quality designs quickly and who want to power their devices with confidence," he adds.

The E36100 series adds five new models with up to 100V or 5A output to Keysight's extensive portfolio of bench power supplies. Its compact form factor (2U, 1/4-rack) saves space on the bench or in the rack, and standard LAN (LXI Core) and USB interfaces make it easy to connect the power supplies to a computer. An intuitive onscreen menu system makes it easy for engineers to perform manual tasks quickly, and over-voltage and over-current detection helps them protect their DUTs.

E36100 series models come standard with measurement capabil-

Different types of bench power supplies

Battery eliminator (B&K 1502). These units are inexpensive and handy when one needs to work on battery-operated equipment as these allow one to work on the equipment without having to find necessary batteries.

Constant-voltage supply (B&K 1686A). These provide adjustable and constant voltage and their maximum output current may be proportional to the output voltage, rather than supplying the rated current at any output voltage.

Constant-voltage/constant-current supply (B&K 1621A). Probably the most popular type of lab power supply is a constant-voltage/constant-current supply. In addition to supplying constant voltage, these also supply constant current. When in constant-current mode, the power supply maintains the set current regardless of changes in the load's resistance.

Multiple-output supply (B&K 9130). Multiple-output power supplies have more than one DC output, often two or three. These are useful and cost-effective for systems that require multiple voltages. An often-used power supply for circuit development is a triple-output supply. One output supplies 0V-6V intended for digital logic. The other two supply (typically) 0V-30V, which can be used with bipolar analogue circuitry.

The supply has a number of useful features. Outputs can be set to timed operation; after a time interval has passed, the output turns itself off. Voltage limits are settable for all channels, so your prototype electrical design can be protected from an accidental over-voltage setting. The two 30V channels can be connected in series or parallel for higher voltage or current, respectively. There are also storage registers for saving up to 50 instrument states for easy recall later (useful for repetitive testing).

Programmable supply. Programmable power supplies are sometimes called system power supplies, as these are often used as part of a computer-operated system for testing or production. Two of the most popular computer interfaces have been GPIB and RS232 serial communications. Ethernet and USB interfaces have also been used. These can be controlled with a computer rather than pushing front-panel keys. This is very useful especially when doing more complex setups like generating dynamic voltage steps using list mode.

Multi-range supply (B&K 9110). Most conventional power supplies operate with fixed voltage and current ratings, for example, 90W/30V/3A. For all other voltage/current combinations, output power will be less. Multi-range supplies are different in that these recalculate voltage/current limits for each setting, forming a constant power hyperbolic-shaped boundary as shown Fig. 1(c). 100W/60V/5A is an example of this type supply. Any voltage/current combinations that lie on the hyperbolic curve are possible, for example, 20V/5A or 60V/1.66A, and in each case, the supply operates at maximum power.

Benefits of this architecture are clear. A multi-range supply offers greater flexibility in output ratings and allows users to substitute several fixed ratings with a single multi-range supply, thus saving cost and bench space.

ity for very small currents, a high-contrast OLED display that makes it easy to view the screen from anywhere, even from sharp angles, support for BenchVue software, IVI drivers and SCPI commands, three-year warranty and Keysight's worldwide support.

PeakTech's newly-developed, linear-regulated dual-channel laboratory power supply (2 x 0-30V/0-5A DC/5V/1A fixed) with a modern design and

the latest technology is provided with a blue LED display for fast and accurate reading of voltage and current values. Output voltage and current limit for the main channels are each continuously variable and can be connected independently from each other in series or parallel mode. Thus, with outputs connected in series, the maximum output voltage rises to 60V DC and outputs connected in parallel double the maximum output current to 10A DC. This device combines easy handling and high efficiency to offer universal applications in research, development, electrical engineering or training. **EFY**



Fig. 3: Typical DIY power supply